COURSE

IB Diploma Programme Business management HL

Break-even analysis

Unit 5: Operations management · 5.5 Break-even analysis

Revision for Unit 5: Operations management: each firm works it out on its own deals.

Class	Topic
IB Diploma Programme Business management HL	**Break-even analysis**

THE LESSON

Setup 2 min

Round 1 Decisions 3 min · Deals 5 min · Topic tasks 4 min

Firm decisions	Deals	TASK 1 Cars to break even	TASK 2 Break-even limits

Round 2 Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min

Firm decisions	Deals	TASK 3 Safety margin as %	TASK 4 Profit on the deal	B2B deal

Round 3 Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min

Firm decisions	Deals	TASK 5 Which price to set	TASK 6 Margin after rise	B2B deal

Final questions 3 min

45-minute setting shown. The 30-minute setting has one topic task per round (Tasks 3, 5 and 6) and one B2B deal. 60 and 75 add more deals.

Each step

Firm decisions	Each role makes its decision on its own page.
Deals	Each pupil trades with a classmate in another firm.
Topic task	The firm answers together. Everyone presses Agree.
B2B deal	One firm sells a lot of cars to another firm.
Final questions	Each pupil answers six questions.

Firm decisions

CEO	**Staff**	Hire staff. Each costs a salary and gives one benefit.
	Sabotage FROM ROUND 2	Hit a rival firm's cash, at a cost. A Legal Counsel blocks it.
Sales & Marketing	**Marketing**	Book one advert campaign to lift the next sales.
Finance	**Capital & Loans**	Borrow cash. The firm pays it back with interest.
	Investments	Buy property or funds. The value can rise or fall.
Operations	**Buying**	Hire the Procurement Specialist or the Stock Controller for the round.

B2B deal · Rounds 2 and 3

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

Topic tasks

Each firm gets its own figures. The method is the same for every firm.

Task 1

ROUND 1

They take up to 5 cars.

Cars to break even? Round up.

WORK OUT THE NUMBER · EXAMPLE FIGURES

Livery, one-off	**£2,400**
Staff training, one-off	**£2,150**
Customer pays per car	**£18,760**
You pay per car	**£16,800**

METHOD

Break-even = (both one-off costs) ÷ (customer pays – you pay), rounded up

(£2,400 + £2,150) ÷ £1,960 = 2.32 → 3 cars

Why: Part of a car does not sell: 3 whole cars cover the costs.

Task 2

ROUND 1

You plan a fleet deal with a break-even chart.

Which two are limits of break-even analysis?

PICK TWO

- ✔ **It assumes every car is sold**
- ✔ **It assumes one fixed price**
- ○ It cannot show the profit
- ○ It leaves out the fixed costs
- ○ It needs the firm's cash balance

Why: It assumes all output sells at one price; it does show fixed costs and profit.

Task 3

ROUND 2

Three orders, each with its break-even quantity.

Order by margin of safety as a %, lowest first?

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **Order 20 cars, break-even 16**
2. **Order 8 cars, break-even 6**
3. **Order 10 cars, break-even 7**

Why: (20 − 16) ÷ 20 = 20%, (8 − 6) ÷ 8 = 25%, (10 − 7) ÷ 10 = 30%.

Task 4

ROUND 2

A deal with a set-up cost.

Profit on the deal?

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

Livery and set-up, one-off	**£6,550**
Customer pays per car	**£12,475**
You pay per car	**£10,900**
They order	**9 cars**

METHOD

Profit = cars ordered × (customer pays − you pay) − set-up

9 × (£12,475 − £10,900) − £6,550 = £7,625

Why: The first cars pay the set-up; the rest is profit: £7,625.

Task 5

ROUND 3

You expect 7 sales at either price. At £9,500 a car you break even at 6; at £9,000, at 8.

Which price should you set, and why?

PICK A OR B AND GIVE A REASON

- ○ £9,000 a car
- ✔ **£9,500 a car**

Why: At 7 sales £9,500 clears its break-even of 6; £9,000 falls short of 8.

Task 6

ROUND 3

They order 8 cars.

Margin of safety now, in cars? Round break-even up.

WORK OUT THE NUMBER · EXAMPLE FIGURES

Livery and set-up, one-off	**£7,900**
Customer pays per car	**£16,665**
Cost per car before the rise	**£13,150**
Cost rise	**10%**

METHOD

Margin of safety = cars ordered − new break-even (cost after the rise)

£13,150 + 10% = £14,465; £7,900 ÷ £2,200 → 4; 8 − 4 = 4

Why: The cost rise ate into the margin of safety: 4 cars left.

Final questions

6
questions for each pupil

109
Level 3 Business questions

8
of them on break-even analysis

The game draws 6 questions on this topic.

1 A component sells for £48. Variable cost is £30 a unit and fixed costs are £54,000 a month. What is monthly break-even output?
3,000 units.

2 A repair plan costs £36 a month, variable cost £12, fixed costs £72,000. Expected sales are 4,000 plans. What is the margin of safety?
1,000 plans.

3 A vehicle sells for £22,500 and its variable acquisition and preparation cost is £17,100. What is unit contribution before fixed costs?
£5,400.

4 Monthly fixed costs rise from £90,000 to £108,000 while contribution remains £300 per vehicle. By how many vehicles does break-even output increase?
60 vehicles.

5 Contribution is £2,400 a vehicle at today's price; matching a rival cuts it to £1,500. Fixed costs are unchanged. Which effect follows?
Break-even rises: each sale covers less fixed cost.

6 A tyre service charges £120 per job, variable cost is £72 and monthly fixed costs are £38,400. What is break-even output?
800 jobs.

7 A service plan sells for £50 a month, has variable fulfilment cost of £18 and fixed monthly support cost of £96,000. What is break-even membership?
3,000 members.

Final questions

8 Fixed costs are £84,000. Each vehicle sells for £16,000 and has a variable cost of £12,000. What is the break-even output?

21 vehicles.